

Kyle Jacob

Senior Data Scientist | Applied ML Engineer

 kylecob67@gmail.com

 +1 (725) 214-2060

 Milford, Connecticut

SUMMARY

Highly skilled and results-oriented **Senior Data Scientist** with 12 years of experience in **machine learning, AI engineering**, and **data science**, specializing in cutting-edge technologies in **generative AI, grid optimization, autonomous systems**, and **data analytics**. Proven track record in leading cross-functional teams to deliver impactful, large-scale solutions for **cloud platforms, robotics, real estate analytics**, and **personalized user experiences**. Expertise in **AWS, TensorFlow, PyTorch**, and **MLOps**, with a passion for building scalable, production-ready models to drive innovation across diverse industries.

SKILLS

Machine Learning & AI Python, TensorFlow, PyTorch, XGBoost, scikit-learn, Keras, Matplotlib, NLTK, Azure Machine Learning	Generative AI & NLP Langchain, LangGraph, AWS Bedrock, OpenSearch, RAG/Graph-RAG, Transformer models, Hugging Face
Deep Learning & Computer Vision CNN, LSTM, Object Detection, Segmentation	Cloud & MLOps AWS (SageMaker, S3, Bedrock), Kubernetes, Docker, CI/CD, Databricks, Spark
Big Data & Engineering Hadoop, Kafka, Snowflake, Redshift, Apache Airflow, Pandas, Numpy	Data Visualization: Pandas, Matplotlib, Seaborn
Data Science & Analytics Time-series forecasting, Anomaly detection, Statistical methods, Optimization algorithms	Programming Languages Python, SQL, Scala, JavaScript, Java, R
Tools & Platforms GitHub, Jira, Tableau, Power BI, Looker, D3.js, Jira	Collaboration & Leadership Cross-functional collaboration, Mentorship, Stakeholder alignment

PROFESSIONAL EXPERIENCE

Collx <i>Senior Data Scientist</i>	12/2023 – present United States
- Leveraging advanced Generative Artificial Intelligence (AI) technologies, with a strong emphasis on Retrieval-Augmented Generation (RAG) architecture, to solve complex business challenges, streamline workflows, and empower data-driven decision making processes. - Exhibiting advanced skills in data analysis and the development of AI/ML models, expertly customized to meet the unique needs of the business. - Designed and implemented real-time data pipelines using AWS Bedrock and Langchain to process multimodal data, streamlining model performance for personalized user recommendations. - Partnering with cross-functional teams to identify and implement optimal AI solutions, leveraging both Generative AI and traditional deep learning methodologies to address diverse business challenges effectively.	

- Applied **Natural Language Processing (NLP)** techniques to develop chatbots and recommendation systems, leading to a 15% increase in conversion rates.
- Built and deployed **real-time predictive models** to optimize platform performance, utilizing **machine learning algorithms** such as **XGBoost** and **scikit-learn**.
- Mentored junior data scientists and guided them in building efficient **MLOps pipelines**, ensuring smooth model deployment and continuous monitoring.
- Acquiring in-depth expertise in **Agile** development methodologies and project management tools, ensuring efficient project execution and the on-time delivery of high-quality solutions.
- Contributing to high-impact Generative AI projects, including developing an **AI-driven application** for insurance eligibility verification and policy recommendations, automating legal document data entry, building a **Knowledge Graph-based RAG application**, and enhancing data query and summarization capabilities for **Databricks tables**
- Enhanced **data infrastructure** by integrating **cloud-based solutions** to process and analyze large datasets, reducing operational costs by 25%.

Nuro

07/2020 – 11/2023 | United States

Senior Data Scientist

- Designed and optimized **computer vision systems** for **autonomous vehicle operations**, improving object detection accuracy by 30% through advanced **deep learning** models.
- Collaborated with robotics engineers to integrate **AI-driven models** for real-time decision-making in autonomous vehicles, enhancing safety features and operational reliability.
- Developed **predictive models** for vehicle maintenance using **time-series analysis**, leading to a 25% reduction in unplanned maintenance events.
- Led the integration of **deep learning** and **reinforcement learning** techniques to optimize route planning and vehicle behavior in dynamic environments.
- Developed and deployed **cloud-based models** using **AWS** services, ensuring real-time data processing and smooth interaction with autonomous systems.
- Conducted **model validation** to ensure the reliability of systems in real-world environments, increasing operational safety by 20%.
- Collaborated with cross-functional teams to develop solutions for **predictive maintenance**, **real-time traffic monitoring**, and **safety enhancements**.

Snowflake

01/2017 – 06/2020 | United States

Senior Machine Learning Engineer

- Led the creation and optimization of **machine learning models** to analyze and derive actionable insights from **large-scale cloud data**, optimizing performance across Snowflake's platform.
- Developed and deployed scalable **ML pipelines** using **AWS SageMaker**, **Databricks**, and **Python**, ensuring efficient and automated workflows for model training, evaluation, and deployment.
- Enhanced data performance and model scalability by integrating **ML models** into Snowflake's cloud platform, improving query processing by 20%.
- Architected and managed **data infrastructure** for high-volume data handling, utilizing **Databricks**, **AWS**, and **Apache Spark** for large-scale machine learning tasks.
- Worked closely with **data engineers** and **cloud architects** to develop solutions that optimized **real-time data processing**, providing Snowflake clients with faster and more accurate insights.
- Led R&D on new **model optimization techniques** and best practices, contributing to Snowflake's innovative approach to **cloud-native ML solutions**.

Vicarious

05/2015 – 12/2016 | United States

Data Scientist

- Designed and implemented **deep learning models** for **robotic perception systems**, enabling real-time object detection and environmental interaction for automation tasks.
- Applied **computer vision** techniques to improve the **accuracy** of robotics vision systems, optimizing object recognition capabilities in dynamic environments.
- Developed and deployed **AI-driven solutions** for robotics systems using **TensorFlow, Keras, and PyTorch**, leading to a 15% increase in the efficiency of automation processes.
- Collaborated with engineers to integrate **vision-based models** into robotics hardware, ensuring seamless data flow and real-time model updates.
- Led the **data preprocessing** efforts for large-scale vision datasets, enhancing the speed and efficiency of the model training process.
- Conducted **extensive testing** and **validation** of AI models to ensure safety and performance in various operational environments.
- Worked closely with hardware engineers to enhance sensor data fusion techniques, leading to more reliable **robot decision-making** in real-world applications.
- Contributed to the development of **active learning** strategies to accelerate training and improve model performance using a smaller subset of data.

Reonomy

07/2013 – 04/2015 | United States

Data Scientist

- Developed **predictive models** for real estate market analysis, providing actionable insights to clients in the commercial real estate sector, driving better investment decisions.
- Designed and implemented **time-series forecasting models** using **XGBoost** and **scikit-learn** to predict real estate trends and property values over time.
- Built and optimized **data pipelines** to handle vast amounts of **geospatial** and **market data**, improving the speed and accuracy of the insights delivered to users.
- Worked with cross-functional teams to integrate **machine learning** models into the **Reonomy platform**, providing clients with predictive analytics for property investment and management.
- Conducted extensive **feature engineering** on large datasets to ensure the accuracy of models, reducing prediction errors by 25%.
- Collaborated with data engineers to optimize **database queries** and streamline data flow, reducing processing time by 20%.

EDUCATION

Princeton University

2009 – 2013 | United States

Bachelor of Science in Computer Science

Relevant Coursework: Machine Learning, Data Science, Cloud Computing, Artificial Intelligence